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Aplicación de métodos de Inteligencia Artificial y Aprendizaje Profundo *(Deep Learning)* al análisis de datos genómicos y transcriptómicos en cáncer

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Aplicación de métodos de Inteligencia Artificial y Aprendizaje Profundo *(Deep Learning)* al análisis de datos genómicos y transcriptómicos en cáncer

Resumen

El cáncer de mama es el tumor maligno más frecuente en mujeres y presenta una notable heterogeneidad molecular que dificulta la estimación de su pronóstico. La disponibilidad de datos transcriptómicos de alta dimensión (RNA-seq) procedentes de iniciativas como *The Cancer Genome Atlas (TCGA)* ofrece la oportunidad para mejorar la predicción de la supervivencia e identificar nuevos marcadores moleculares mediante el aprendizaje automático *(Machine Learning)*.

El principal objetivo de este trabajo es la mejora de la predicción de pronóstico y supervivencia utilizando modelos "deep learning" y la identificación de marcadores moleculares asociados a pronóstico utilizando dichos modelos. Para ello se ha realizado un estudio riguroso de cinco métodos, de análisis de supervivencia. (regresión de Cox penalizada (Lasso, Elastic Net y Ridge), Random Survival Forest y dos redes neuronales profundas (DeepSurv y Cox-nnet)) sobre datos de RNA-seq de 519 pacientes de cáncer de mama del *TCGA*, con la supervivencia global truncada administrativamente a 60 meses.

Se diseñó una metodología para estar libre de fuga de información (information leakage): la selección de genes y el ajuste de hiperparámetros en los modelos se realizan exclusivamente sobre el conjunto de entrenamiento, reservando un conjunto de prueba independiente para la validación de los resultados experimentales.

Los modelos se compararon mediante el índice de concordancia (**C-index)** y la prueba de **log-rank** sobre curvas de **Kaplan–Meier** estratificadas por el riesgo predicho. El modelo profundo **DeepSurv** obtuvo el mejor poder discriminante (C-index en torno a 0,82) y la única estratificación de riesgo estadísticamente significativa, superando a los modelos lineales de Cox y al Random Survival Forest. Por último, se emplearon SHAP y análisis de enriquecimiento funcional para interpretar los modelos e identificar firmas génicas asociadas a pronóstico en cancer de mama con sentido biológico.

Abstract

Breast cancer is the most common malignant tumor among women and exhibits considerable molecular heterogeneity, making prognosis estimation a challenging task. The availability of high-dimensional transcriptomic data (RNA-seq) from initiatives such as the *The Cancer Genome Atlas* provides an opportunity to improve survival prediction through machine learning techniques.

The objective of this Bachelor's Thesis is to compare six survival analysis models (penalized Cox regression (Lasso, Elastic Net, and Ridge), Random Survival Forest, and two deep neural network approaches (DeepSurv and Cox-nnet)) using RNA-seq data from 519 breast cancer patients from TCGA, with overall survival administratively censored at 60 months.

A methodology designed to prevent information leakage was implemented, where gene selection (limma and univariate Cox-based gene selection) and hyperparameter tuning (cross-validation) were performed exclusively on the training set, while the test set was reserved for a single final evaluation.

Model performance was assessed using the concordance index **(C-index)** and the **log-rank** test on **Kaplan–Meier survival** curves stratified by predicted risk. The deep learning model **DeepSurv** achieved the best discriminative performance (C-index of approximately *0.82*) and was the only model to produce a statistically significant risk stratification, outperforming both the linear Cox models and the **Random Survival Forest.**

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Descriptores

Palabras clave: Análisis de supervivencia; Aprendizaje profundo; Transcriptómica (RNA-seq); Pronóstico del cáncer de mama; Inteligencia artificial explicable (SHAP).

Keywords: Survival analysis; Deep learning; RNA-seq transcriptomics; Breast cancer prognosis; Explainable AI (SHAP).

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1. Introduction

1.1 Context and motivation

Cancer remains a leading cause of death worldwide, and breast cancer is the most frequently diagnosed cancer in women. Beyond its incidence, breast cancer is not a single disease, but rather a collection of molecularly dysfunctional entities. Because the tumours look similar under the microscope and share the same clinical stage can follow very different clinical conclusions, which makes the estimation of a patient prognosis a genuinely difficult task. Molecular taxonomy has refined this picture by distinguishing intrinsic subtypes, namely Luminal A, Luminal B, HER2-enriched and triple-negativem (TNC) breast cancer, each associated with a different biology and outcome. This problem motivates the search for prognostic models that go beyond classical clinical variables and exploit the molecular profile of each type of tumour.

RNA sequencing (RNA-seq) is the expression of tens of thousands of genes simultaneously, and large public societies such as *The Cancer Genome Atlas (TCGA)* have released thousands of tumour samples with gene-expression and clinical follow-up data. These resources make it possible to approach cancer prognosis as a data-driven prediction problem: given the data of a tumour, can we predict how long a patient is likely to survive? Answering this question is crucial for oncologists to stratify patient groups based on clinical markers into risk subgroups, based on molecular profiles , and thereby be able to offer personalised treatments.

Answering this question raises two challenges.

First: survival data are censored, since for many patients we only know that they were alive up to a certain time, so standard regression or classification cannot be applied directly because of that.

Second: the data are extremely high-dimensional, with far more genes than patients, which makes models prone to overfitting.

The tool for censored outcomes is the Cox proportional-hazards model [4], is a linear model of the log-hazard to predict the hazard(risk) for the patients. In this high-dimensional setting it must be regularized because it is very prone to overfitting, giving rise to penalized variants such as *Lasso*, *Ridge* and *Elastic Net* that perform feature selection.

Tree-based ensembles such as the *Random Survival Forest* [7] the linearity assumption and can capture interactions between genes in a nonlinear aspect.

More recently, Deep learning has been adapted to survival analysis: DeepSurv [9] replaces the linear predictor of the Cox model with a neural network, and Cox-nnet [3] follows a closely related idea, both optimizing the Cox partial likelihood. These models can, in principle, learn the

non-linear relationships that link gene expression to risk, but they are often criticised as black-box methods whose predictions are hard to justify in a clinical context.

There are two main problems that motivated this work. The first problem is ensuring that the methods used are fair and reliable. When researchers compare models that predict patient survival, they often make a mistake by using information from the test data to select features or tune model parameters. This can make the results appear better than they actually are. To ensure a fair comparison, all decisions related to feature selection and model tuning should be made using only the training data.

The second problem is understanding how the models make their predictions. It is not enough for a model to achieve good predictive performance. It should also provide insight into which genes are most important for the prediction process. This is necessary to connect the results with existing biological knowledge and to increase confidence in the findings. Techniques such as SHAP can help explain model predictions, even when the models are highly complex. As a result, these models do not have to be treated as complete "black boxes" whose decisions cannot be understood.

This project aims to address both of these problems. It develops a pipeline that is both fair and transparent. The pipeline compares three types of survival models using breast cancer data: penalized Cox regression, Random Survival Forest, and deep neural networks. In addition, the project investigates how the deep learning models make their predictions. The goal is to determine whether deep learning methods provide a real advantage for predicting survival in breast cancer patients. The project does not aim to develop a new algorithm. Instead, it focuses on comparing existing models in a way that is both rigorous and easy to interpret.

1.2 Objectives

The main goal of the project is to investigate the use of deep learning networks to improve prognosis and survival predictions and to identify new molecular markers in breast cancer. Furthermore, a key objective for biologists is to ensure interpretability in terms of the underlying molecular biology. The project analyses recently proposed deep learning models in comparison with benchmark machine learning techniques, both linear (Cox regression) and non-linear (Random Survival Forest).

This general goal is broken down into the following specific objectives:

1. To build a genomic expression dataset for breast cancer using the TCGA database. The resulting dataset must be clean, free from noise, standardised and contain clinical and survival data in order to apply statistical and machine learning techniques
2. Reduce the number of candidate genes using differential expression analysis and univariate Cox regression to keep the computational cost manageable and to remove noise
3. Train and compare five survival prediction models: Cox Lasso, Cox Elastic Net, Random Survival Forest, DeepSurv, and Cox-nnet.
4. Evaluate model performance using Harrell's concordance index (C-index) to measure the ability of the models to rank patients according to survival risk.

5. Assess risk stratification performance by dividing patients into risk groups and comparing their survival distributions using Kaplan–Meier curves and the log-rank test.
6. Apply SHAP-based interpretability methods to the deep learning models in order to identify the genes that contribute most strongly to model predictions.
7. Use bootstrap resampling to evaluate the stability of SHAP-derived gene rankings and generate a consensus list of prognostic genes.
8. Perform functional enrichment analysis on the selected genes to investigate their biological significance and their association with breast cancer-related pathways and processes.
9. Perform an interaction evaluation between the trained model as Random Survival Forest with Minimal Depth and the neuronal-network Cox-Nnet with the STRING database

1.3 Structure of the document

The remainder of this document is organised as follows. Section 2 reviews the state of the art (SOTA), to take a look at what has been used on cancer genomics and RNA-seq. The fundamentals of survival analysis, the machine learning and deep-learning methods used for survival prediction, and the techniques available for model interpretability. Section 3 describes the materials and methods: the TCGA dataset, the preprocessing and feature-selection pipeline, the six survival models, and the evaluation and validation strategy. Section 4 presents the experimental results, including the tables of comparison of the models, the risk-stratification analysis and the interpretation of the deep models through SHAP and enrichment analysis. section 5 discusses the findings, their limitations and their biological implications, and Chapter 6 summarises the main conclusions and outlines possible lines of future work.

2. State of the art (SOTA)

This section reviews the scientific background on the previous investigation . It first introduces cancer genomics and RNA-seq data, then the fundamentals of survival analysis, the machine learning and deep learning methods used to predict survival, and finally the techniques available to interpret these models.

2.1 Cancer genomics and RNA-seq

A tumor's biological identity is largely dictated by its active genes. According to the central dogma of molecular biology, genetic information flows from DNA to RNA, and is ultimately translated into proteins. Measuring messenger RNA (mRNA) abundance gives us a precise, quantitative snapshot of gene activity within a biological sample.

While researchers traditionally relied on microarrays for this task, the scientific field has largely transitioned to RNA sequencing (RNA-seq). By directly counting the sequence reads that map to each transcript, RNA-seq offers a vastly superior dynamic range without being limited by predefined probes. For this project, expression values are summarized using the RSEM method the standard for quantifying TCGA RNA-seq data.

Much of this breakthrough work has been done by *The Cancer Genome Atlas (TCGA)*, an initiative dedicated to mapping human tumors at the molecular level. Their breast cancer working group analyzed data from over a thousand patients, showing copy-number variations, mutations, and expression profiles..

The dataset utilized in this study was sourced from the *cBioPortal* [21] *for Cancer Genomics.* Originally introduced by [1], cBioPortal acts as an open source platform that has cancer genomics data, allowing researchers to easily explore, analyze, and have insights from massive datasets like TCGA.

However, analyzing gene-expression comes with unique computational cost. We typically face a classic "high-dimensional" bottleneck where the number of measured genes outnumbers the available patient. This regime heavily favors overfitting over the models.

Furthermore, a large portion of these genes exhibit zero or trivial expression levels across most samples. This high prevalence of near-zero counts in RNA-seq data arises from both biological realities, where many genes are simply inactive or tightly repressed in specific cell types and technical limitations, such as transcript "dropouts," where rare RNA molecules are missed during the sequencing sampling process due to limited sequencing depth. To address this and reduce statistical noise, the dataset is filtered to eliminate low-expression genes with a threshold of 0.8, narrowing it down to a highly informative subset for differential expression analysis. For this

crucial step, the limma package remains the standard, leveraging empirical Bayes moderation to ensure robust and reliable statistics even with limited sample sizes [15].

2.2 Fundamentals of survival analysis

Survival analysis models the time until an event of interest, such as death or relapse happens. Its defining feature is censoring: for many patients the event has not been observed by the end of follow-up, so the only information available is that they were event-free up to a certain time, and simply discarding these observations biases the analysis. The two central functions are the survival function, the probability of surviving beyond a given time, and the hazard function, the instantaneous risk at each instant of an event. The survival function is estimated with the Kaplan–Meier estimator [8], and two groups of patients are compared with the log-rank test, the ones with a median above the mean "*the risk group*" and below the median "*the low-risk group*".

The reference model is the Cox proportional hazard model [4], a linear model that writes the hazard as a baseline hazard multiplied by the exponential of a linear combination of the covariates and is fitted by maximising the partial likelihood, avoiding any assumption about the shape of the baseline hazard. Its key assumption is that hazard ratios between patients are constant over time. The discriminative ability of a survival model is usually summarised by concordance index, or C-index [6]: the probability that, for a comparable pair of patients, the model assigns the higher risk to the one who experiences the event first, where 0.5 denotes random ordering and 1.0 perfect discrimination.

When the covariates are numerous, as with gene expression, the Cox model must be regularized. The Lasso adds an L1 penalty that drives many coefficients to exactly zero and therefore selects features;

Ridge adds an L2 penalty that shrinks them without removing them; and the Elastic Net combines both penalties and tends to retain groups of correlated genes [18]. Efficient coordinate-descent algorithms fit the whole regularization path of these penalized Cox models [16], and ready-to-use implementations are available in libraries such as scikit-survival [14], which is used in this work.

2.3 Machine and deep learning for survival prediction

Penalized Cox models have an important limitation: because they model the log-hazard as a linear combination of covariates, they cannot capture non-linear relationships or interactions between genes and survival unless these are explicitly specified through interaction terms or basis expansions.

Tree-based methods overcome this constraint. The Random Survival Forest (RSF), for instance, grows an ensemble of survival trees. At each node, it selects the split that maximizes the difference in survival experience between the resulting groups (typically via a log-rank splitting rule), then aggregates predictions across many such trees. Because splits are chosen adaptively from the data, RSF can capture non-linear effects and gene–gene interactions without requiring them to be specified in advance.

A more recent line of work extends the Cox model using neural networks. DeepSurv [9] keeps the Cox partial-likelihood as its loss function but replaces the linear risk predictor with a multilayer perceptron, so the log-risk becomes a **non-linear** function of the input covariates. Cox-nnet [3] follows a similar strategy but is built specifically for high-dimensional omics data, and additionally provides gene and pathway-level importance scores derived from the trained network.

The pycox library [10] implements these and related neural extensions of the Cox model; the accompanying paper discusses several ways of increasing the flexibility of neural survival models. Greater flexibility, however, does not guarantee better predictive performance — several benchmarking studies show that for tabular and omics data with small sample sizes, deep survival models do not consistently outperform a well-regularized Cox model.

Because benchmark results can vary substantially depending on how the comparison is designed (data-splitting strategy, hyperparameter tuning, choice of evaluation metric), a rigorous, standardized evaluation protocol is essential for a fair comparison. This is the purpose of the evaluation protocol used in this project: to compare penalized Cox regression, Random Survival Forests, and neural Cox-based models (DeepSurv, Cox-nnet) under consistent, unbiased conditions to determine which approach performs best for this specific dataset and task.

2.4 Differential Expression and Limma

Differential expression analysis is historically the most common approach to identify genes associated with a clinical outcome in transcriptomic studies. This approach is based on a statistical contrast between the expression level of each gene across two or more groups for example, high-risk versus low-risk patients defined from a survival model in order to determine whether the observed difference is greater than what would be expected by chance. Formally, a contrast is a linear combination of the coefficients of a model that represents the specific comparison of interest, and it is tested against the null hypothesis that this difference is equal to zero. Applying this contrast independently to every gene in the dataset produces, for each gene, a p-value typically adjusted for multiple testing using methods such as Benjamini-Hochberg together with a log fold change that reflects the magnitude of the difference, allowing a list of differentially expressed genes (DEGs) to be built between the groups under comparison.

One of the most widely used tools to carry out this type of analysis is limma (Linear Models for Microarray Data), an R/Bioconductor package that fits a linear model to the expression values of each gene and defines the contrasts of interest through a design matrix and a contrast matrix [15]. Its main contribution is an empirical Bayes method that moderates the variance estimate of each gene by borrowing information across all genes in the dataset, instead of estimating it independently for each one. This is particularly useful when sample sizes are small, as is common in gene expression studies, since it stabilizes variance estimates, increases the statistical power of the contrast and reduces the risk of false positives compared with independent gene-by-gene t-tests. Although limma was originally developed for microarray data, the voom transformation extends its use to RNA-seq data by converting raw read counts into log-CPM values and modeling the mean-variance relationship to assign a precision weight to each observation [11], which has made limma one of the reference tools for differential expression analysis.

2.5 Model interpretability in biomedicine

In real-life applications a model's performance is not enough; it must also be understandable for validation, trust and biological insight.

Penalized Cox models are easy to understand because the size and direction of each coefficient directly show the effect of a gene on the hazard. This makes sense because you can see how each gene affects the outcome.

More complex models like Random Survival Forests and deep neural networks need extra steps to understand what drives their predictions. One popular approach is SHAP, which explains how much each input feature contributes to a prediction. SHAP works with both tree-based models and neural networks, giving a measure of feature importance.

SHAP is based on Shapley values, a concept from cooperative game theory originally designed to fairly distribute a payout among players according to their contribution. In this context, each feature is treated as a "player" and the model's prediction as the "payout". For a given sample, SHAP estimates each feature's contribution by comparing the model's prediction with and without that feature, averaged across many possible combinations of the other features. This makes sense because it captures not only the isolated effect of a gene but also how its effect changes depending on which other genes are present in the model.

However, when working with biological datasets, feature rankings may not be stable due to sampling variability repeatedly identified as relevant across resampled datasets.

Finding genes is just the first step; these findings must then be linked to existing biological knowledge. Functional enrichment analysis helps by testing whether a set of selected genes is overrepresented in known biological processes, molecular functions, cellular components or signaling pathways.

This analysis often uses curated databases like Gene Ontology [19] and Reactome [20] . It can be performed using tools like Enrichr [2] and gseapy. By mapping gene lists to functions and pathways, enrichment analysis turns statistical findings into biologically meaningful hypotheses and helps interpret prognostic signatures.

This step is crucial for understanding the significance of the results. This involves retaining those genes that are repeatedly identified as relevant across resampled datasets.

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3. Requirements specification

3.1 Scope and nature of the system

This project is focused on developing a tool for researchers. It is not intended for general users, but for people who want to analyze breast cancer data and study factors related to patient survival. The tool processes large amounts of transcriptomic and clinical data, builds predictive models, and presents the results in a way that is easier to understand and interpret.

The requirements of the system are listed below. They were defined according to the objectives established at the beginning of the project and describe the main capabilities that the tool must provide.

For clarity, the requirements have been divided into four groups:

- functional requirements: which describe what the system must do
- operational requirements: which describe how the system should work
- data requirements: which define the characteristics of the data used by the tool
- integration requirements: which specify how the system interacts with external resources and software components.

3.2 Functional requirements

The pipeline must provide the following functions:

1. Acquire the TCGA breast-cancer RNA-seq (RSEM) expression matrix and the associated clinical table obtained through *cBioPortal* [21].

2. Assign each patient a molecular subtype (*Luminal A, Luminal B, HER2-enriched or TNBC*) from the ER, PR and HER2 markers, discarding cases that cannot be classified.

3. Filter out genes expressed at zero in more than 80% of the samples.

4. Define the overall-survival outcome and administratively truncate it at 60 months, producing the truncated time and event labels.

5. Partition the patients into training and test sets (and into train/validation/test for the non linear neural models) before any feature selection, using a fixed partition that is persisted and shared across all experiments.

6. Select genes by limma differential expression followed by a univariate Cox screen, computed exclusively on the training split, and apply the resulting panel to both splits.

7. Train the six survival models: Cox Lasso, Cox Elastic Net, Cox Ridge, Random Survival Forest, DeepSurv and Cox-nnet.

8. Select hyperparameters without using the test set: Coxnet penalties and Random Survival Forest parameters by k-fold cross-validation on the training set, and the neural-network architecture on a validation set.

9. Evaluate every model on the untouched test set with the concordance index, and stratify patients by median predicted risk to produce Kaplan–Meier curves and a log-rank test.

10. Estimate SHAP feature attributions for the neural models with bootstrap resampling and derive a consensus ranking of prognostic genes.

11. Test the selected genes for over-representation in Gene Ontology and Reactome gene sets.

12. Export the model-comparison table, the selected gene panels, the bootstrap statistics and the figures used in the report.

3.3 Non-functional requirements

1. Reproducibility: every stochastic step must run under a fixed random seed 42, and the patient partition must be stored and reused.

2. Absence of data leakage: no information from the test set may influence preprocessing, feature selection or hyperparameter tuning.

3. Modularity and maintainability: reusable preprocessing, modelling and evaluation code must be kept separate from the exploratory notebooks.

4. Transparency: every reported number must come from a single, traceable model configuration, and the predictions of the deep models must be explainable.

5. Computational feasibility: the pipeline must run on a single workstation without specialised hardware (*GPU's)*.

6. Use of open, standard tools: the system must rely on well-established open-source scientific libraries to ease reproduction.

3.4 Data and information requirements

The system consumes two primary inputs: a gene-expression matrix (genes by samples, with RSEM values, file *data_mrna_seq_v2_rsem*) and a clinical table that contains, among other fields, the overall-survival time and status and the ER, PR and HER2 markers needed for subtyping.

Internally it produces a merged sample-by-gene table augmented with the survival labels and the molecular subtype, a persisted train/validation/test partition (splits.json) and the selected gene panels. The outputs are the model-comparison table (concordance index and log-rank p-value), the per-gene bootstrap SHAP statistics and the figures of the report.

3.5 Interfaces

The pipeline is exposed through the input data files described above and a programmatic interface composed of Python modules, mainly accessed through Jupyter notebooks, where the research was carried out. The outputs of the research are tables, CSV files, and figures that present the results obtained from the analyses.

4. Development

4.1 Development methodology

The research project follows an exploratory methodology well suited to research. Each idea was first prototyped in a Jupyter notebook in the research/ directory and, once stable, the reusable logic was modularized into Python modules under a src/ package. An iteration was a methodology of the pipeline: an early version performed feature selection on the whole dataset and selected some hyperparameters on the test set, both of which inflate the apparent performance. These issues were identified and corrected so that all selection now happens on the training data only.

4.2 Technologies and tools

The system is written in Python and its dependencies are managed reproducibly with the *uv package* manager and a pyproject.toml / uv.lock pair. The main libraries are:

Pandas: A data analysis and manipulation library used to work with structured datasets, tables, and data frames.

NumPy: A numerical computing library that provides efficient multidimensional arrays and mathematical operations for scientific computing.

InMoose: A bioinformatics library for analyzing biology data, including gene expression and high-throughput biological datasets.

Limma: A Bioconductor package widely used for differential gene expression analysis in microarray and RNA-sequencing studies.

Scikit-learn: A machine learning library that provides algorithms for classification, regression, clustering, model evaluation, and data preprocessing.

Scikit-survival: An extension of scikit-learn designed for survival analysis.

PyTorch: A deep learning framework used for building, training, and deploying neural networks and other machine learning models.

PyCox: A Python library for deep learning based survival analysis, built on top of PyTorch and designed for time-to-event prediction tasks.

SHAP: An explainable AI library that interprets machine learning models by quantifying the contribution of each feature to a prediction.

GSEApy: A Python implementation of Gene Set Enrichment Analysis (GSEA) used to identify biological pathways and gene sets associated with experimental results.

4.3 System architecture

The code is organised into clearly separated layers.

- **Data:** It's the data directory that has the raw inputs from the dataset

- **Src/dataset:** It's the package that builds the modeling of separation of the dataset
    - **TorchPreprocessing:** it's a class for all the preprocessing to enable the usage of torch
    - **Split_dataset_by_ids:** It's a function in for splitting the dataset in train/validation/test make the splits and leave that splits in ***splits.json.***

- **Src/utils**: A package that contains code for the preprocessing, models and plots:
    - **Preprocessing**: subtype classification, low-expression filtering and limma differential expression
    - **Cox_models:** the Cox fits, the cross-validated selection of Coxnet and Random Survival Forest hyperparameters and the per-gene concordance index
    - **Kapplan_meir**: The Kaplan-Meier curves and the log rank tests.
    - **Set_seed:** For the general seeding of the project for reproducibility
    - **Plots:** Contains all the code for making the plots of the research project

- Research/notebooks: holds the experiments for the research
    - **Limma:** Penalized Cox models and the Random Survival Forest
    - **Nn_Cox_Limma:** Contains the DeepSurv and Cox-nnet
    - **Shap_nn:** the Shap and enrichment

4.4 Implementation of the pipeline

The preprocessing stage begins by assigning a molecular subtype to each patient. This is done using a classifier based on the ER, PR, and HER2 markers. Four molecular subtypes are retained, while patients with unclear classifications are excluded from the analysis.

The gene expression data is then combined with the corresponding clinical information. Genes that are not expressed in a sufficient number of samples are removed, and overall survival times are administratively censored at 60 months.

To identify differentially expressed genes, expression values are transformed to a log2 scale and a design matrix containing the molecular subtype indicators is created. Differential expression analysis is then performed using the limma package.

The dataset is divided into training and test cohorts, and the split is saved to ensure that all analyses use the same patient partitions. Feature selection is carried out using a combination of

limma-based ranking and univariate Cox screening, and the selected genes are subsequently used for all predictive models.

Several survival prediction models are trained, including penalized Cox regression models (Lasso and Elastic Net), Ridge regression, and Random Survival Forest. Model performance is evaluated on the independent test cohort using survival prediction metrics.

Patients are stratified into high-risk and low-risk groups according to their predicted risk scores. Kaplan–Meier survival curves are generated for these groups and compared using the log-rank test.

For the best-performing models, SHAP values are computed to estimate feature importance. The stability of these attributions is assessed through bootstrap resampling, producing a consensus ranking of important genes. Finally, the resulting gene signatures are interpreted through functional enrichment analysis using Gene Ontology and Reactome databases.

4.5 Reproducibility and good practices

Reproducibility its treated as a single set_seed function that fixes the seeds of Python, NumPy and PyTorch. The train/validation/test partition is persisted and shared between notebooks with the *splits.json*; feature selection and hyperparameter tuning are confined to the training data; and the neural networks are switched to evaluation mode before their risk scores are computed

5. Methodology

5.1 Dataset

The dataset used in this study originates from the **Cancer Genome Atlas (TCGA)** breast cancer RNA-seq (RSEM-normalized) gene expression repository. This dataset is widely used in computational oncology due to its large-scale nature and high-quality genomic and clinical annotations, making it particularly suitable for survival and risk prediction tasks. The raw data is structured as a gene expression matrix, where each row corresponds to an individual patient sample (n = 819), and each column represents a gene (n = 20,440). The values in the matrix correspond to normalized gene expression levels (RSEM), which quantify the transcriptional activity of each gene for each patient.

Category / Variable	Number of Patients	Percentage (%)
Age Stratification		
Under 50 years (< 50)	237	28.9%
Between 50 and 60 years ($50 - 60$)	223	27.2%
Over 60 years (> 60)	359	43.8%
Race / Ethnicity		
White	642	78.4%
Black or African American	140	17.1%
Asian	33	4.0%
Not reported / Unknown	4	0.5%
Pathological Stage		
Stage I	142	17.3%
Stage II	473	57.8%
Stage III	181	22.1%
Stage IV	15	1.8%
Not available / Unknown	8	1.0%

Table 1: Demographics and clinical characteristics of the TCGA-BRCA dataset (n = 819)

5.2 Exploratory Data Analysis and Visualization

To study the dataset used in this research project, plotting was done to see if the dataset is sequentially good and every distribution was normalized or not, first it was started with a box plots of the of 20 patients per each class of Luminal A, Luminal B, HER2-enriched and TNBC as explained in the Data preprocessing section.

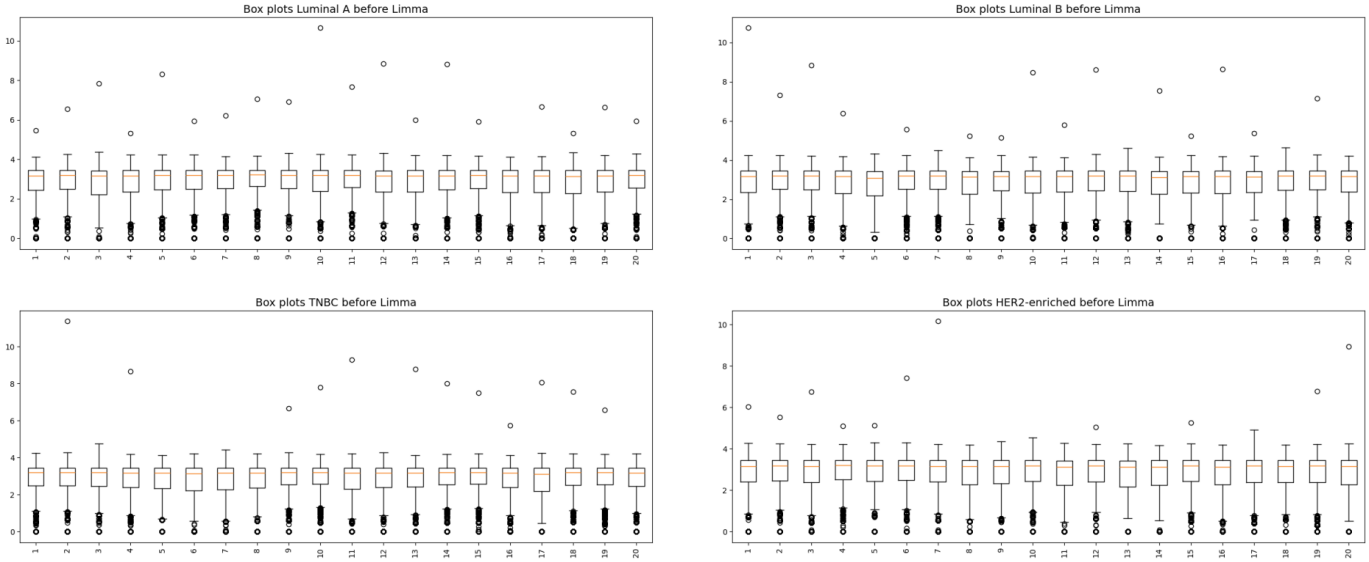


Figure 1: Box-plots of the four molecular subtypes

Figure 1 displays the pre-limma gene expression distributions for 20 representative patients across the four molecular subtypes. While the median expression levels remain stabilized and consistent across all samples, the lower bound of each distribution exhibits a high density of outlier data points approaching zero. These outliers demonstrate that a significant fraction of the genomic profile consists of unexpressed or lowly expressed genes within individual patients. This widespread characteristic of transcriptomic datasets validates our quality control filtering stage, where low-variance and highly sparse genes (>80% zero values) were systematically discarded.

To further investigate the transcriptomic landscape, kernel density plots were generated for each molecular subtype, as shown in **Figure 2**. This visualization technique provides additional insights into the continuous distribution of gene expression levels across all patients, corroborating the sparsity patterns observed in the previous box plot analysis **Figure 1**.

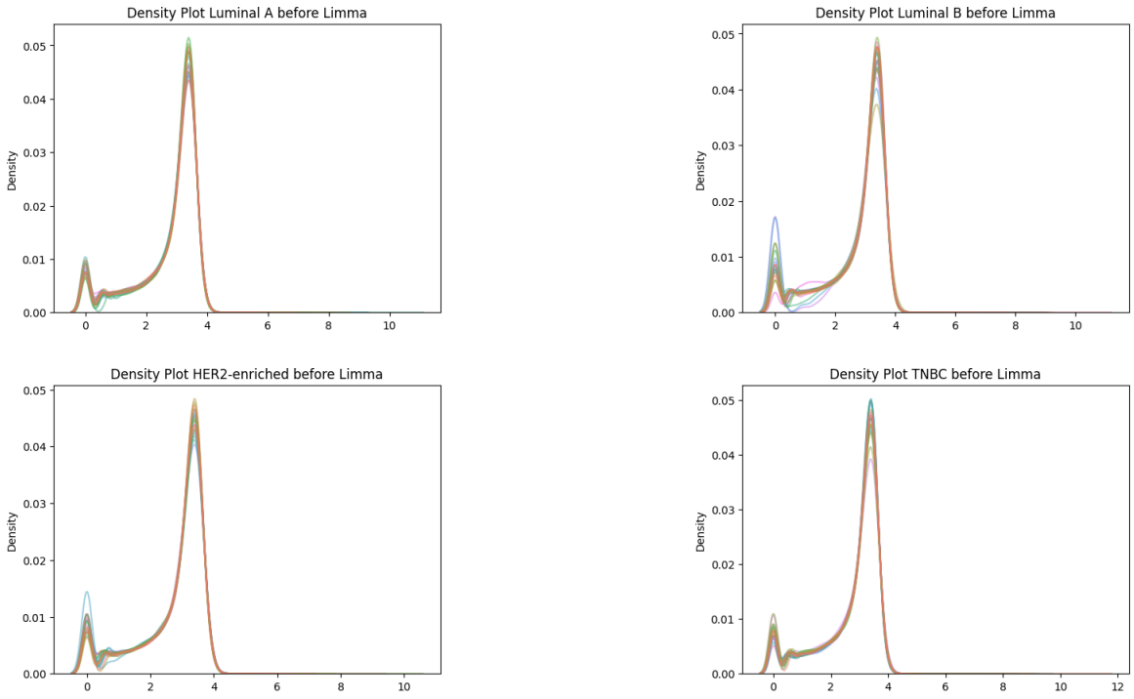


Figure 2: The density plots of the four molecular subtypes

The density profiles across all four breast cancer subtypes reveal a probability distribution with two modes. The primary and highest peak is consistently located between the expression values of 2 and 4, representing the baseline distribution. However, a secondary peak emerges precisely at 0. This substantial accumulation of density at the absolute lower bound confirms that a large fraction of the genomic profile consists of genes with zero or near-zero expression values. This visual evidence of zero-inflation further justifies the implementation of our quality control filtering threshold to remove uninformative, low-expression features before model training.

To conclude the exploratory data analysis, a Principal Component Analysis (PCA) was performed to evaluate the global variance and spatial distribution of the four breast cancer molecular subtypes. As illustrated in **Figure 3**, the pairwise scatter plots of the first four principal components (PC1 to PC4) reveal a highly overlapping distribution among the Luminal A, Luminal B, HER2-enriched, and TNBC samples.

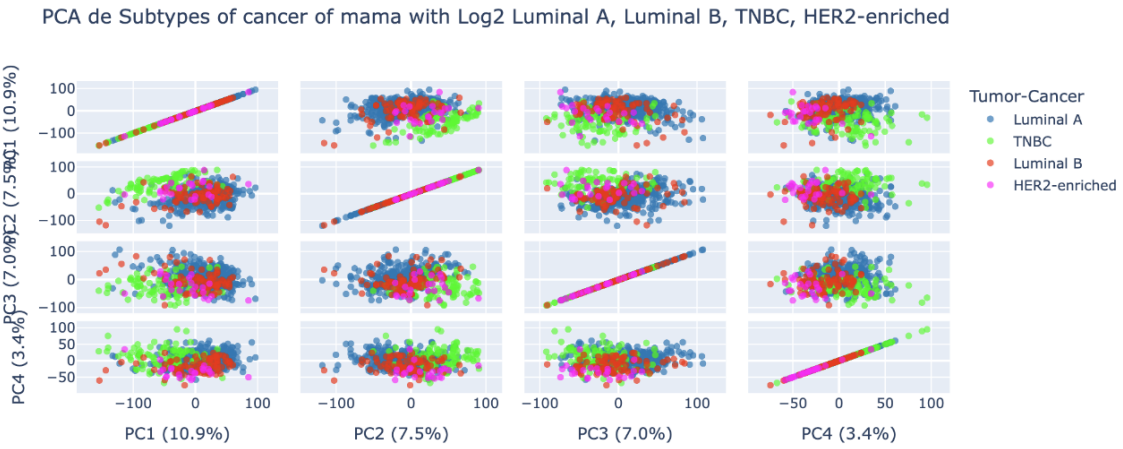


Figure 3: The Principal component analysis (PCA) of the four molecular subtypes

This substantial overlap is a positive indicator of data quality, as it suggests the absence of significant batch effects or systemic technical biases across the dataset. Because the molecular subtypes do not naturally separate along these primary axes of global variance, this analysis validates our methodological choice to apply *Limma* to extract a refined panel of differentially expressed genes, which is necessary to effectively train our downstream survival models.

In summary, the exploratory data analysis shows a high level of dataset sparsity. This is marked by a large number of zero-value outliers. It indicates that many genes in the genomic profile are not expressed. As a result, a thorough feature reduction step is necessary to remove these genes that do not provide useful biological information. Without this filtering, the noise in the raw dataset could undermine the stability and accuracy of our survival models.

5.3 Data Preprocessing

Gene Filtering

The initial dataset consisted of 819 patients and 20,440 genes, categorized into five molecular subtypes with the following baseline distribution:

- **Luminal A:** 330 patients (40%)
- **Luminal B:** 81 patients (10%)
- **HER2-enriched:** 23 patients (3%)
- **Triple-Negative Breast Cancer (TNBC):** 85 patients (10%)
- **Unknown (UNK):** 299 patients (37%)

To ensure the integrity of the downstream analysis, patients classified as "Unknown" (UNK) were excluded. This step removed 37% of the data, resulting in a finalized cohort of 519 patients representing the four primary breast cancer subtypes.

Feature Selection and Quality Control Filtering

To optimize computational efficiency and ensure data quality, the feature space was filtered in two stages. First, a preliminary reduction removed a small subset of genes, adjusting the profile from 20,440 to 20,393 genes. Second, a sparsity threshold was applied to eliminate low-expression features; genes with more than 80% zero values across samples were discarded. This filter eliminated 1,898 genes, leaving a total of 18,492 genes. Due to missing or insufficient data after this thresholding step, 104 patients were also excluded from the cohort.

```python
def eliminate_zero_genes(self, df: pd.DataFrame, column: str, threshold: float = 0.8) -> pd.DataFrame:
    genes = [c for c in df.columns if c != column]
    expression = df[genes]

    zeros_per_gene = (expression == 0).sum(axis=0)
    max_zeros_allowed = len(df) * threshold

    print(f"Genes before Treshold: {len(genes)}")
    print(zeros_per_gene.describe())
    print(f"Threshold (>{threshold*100:.0f}% zeros): {max_zeros_allowed:.0f} samples")

    keep_genes = zeros_per_gene[zeros_per_gene <= max_zeros_allowed].index
    df_filtered = pd.concat([df[column], df[keep_genes]], axis=1)

    print(f"After the treshold: {len(keep_genes)}")
    return df_filtered
```

Figure 4: Code for the elimination of the zeros genes or low expression genes in the dataset

Clinical Follow-up and Event Truncation

For survival and event-time analysis, the clinical follow-up timeline was truncated from 250 months to 60 months. Preliminary experimentation indicated that capping the event window at 5 years (60 months) minimized long-term censoring noise and provided a more robust, statistically accurate distribution for modeling.

Differential Gene Expression Analysis and Feature Selection via Limma

To identify the key genetic associated with different breast cancer profiles, we utilized **limma**, a highly established statistical package designed for assessing differential expression in transcriptomic data introduced in section 2.4 differential expression. Limma relies on linear models and empirical Bayes algorithms, which allows it to generate stable and robust statistical inferences even across complex high-dimensional datasets.

In our workflow, limma was initialized using the training cohort (*df_train*). To prevent confounding variables during the linear modeling phase, downstream survival indicators, specifically *event, event_60*, and *time_60* were explicitly excluded from the expression matrix. The model's design matrix (*design*) was constructed around the primary *Tumor-Cancer* target column to isolate the genetic variance directly attributable to the tumor subtype.

Following the empirical Bayes statistical ranking, we selected the top 1,000 genes showing the most significant differential expression. This subsetting step was deliberately implemented to reduce computational overhead and accelerate the training of subsequent machine learning architectures**.**

The final output was a refined expression matrix (*expr*) consisting of 1,000 highly informative features, ensuring a balance between biological relevance and computational efficiency.

```python
def initialize_limma(self, df : pd.DataFrame, column : str , column_event : str, column_status : str) -> Tuple[pd.DataFrame, pd.DataFrame, np.ndarray]:

    metadata = pd.DataFrame(df[[column, column_event, column_status]], index=df.index)

    metadata.columns = [column, column_event, column_status]

    number_data = df.drop(columns=[column, column_event, column_status])

    number_data = number_data.apply(pd.to_numeric, errors='coerce')

    number_data = np.log2(number_data + 1)

    expr = number_data.T  #(Samples, Genes)

    assert (expr.columns == metadata.index).all() # type: ignore

    metadata_aligned = metadata.loc[expr.columns].copy() # type: ignore

    design = pd.get_dummies(metadata_aligned[column], drop_first=True).astype(float)

    #This is for fitting the models
    limma_fit_models = limma.lmFit(obj=expr, design=design)

    #Emperical moderate Bayes (eBayes)
    limma_fit_models = limma.eBayes(limma_fit_models)

    #Obtain the table of Results
    results = limma.topTable(limma_fit_models, number=np.inf) # type: ignore

    #Transform to pandas dataframe

    results_df = pd.DataFrame(results)


    if column == "Tumor-Cancer":
        results_df = results_df.rename(columns={
                            "column0":"HER2-enriched",
                            "column1":"Luminal A",
                            "column2":"Luminal B",
                            "column3":"TNBC"
                            })

    return (results_df, design, expr)
```

Figure 5: Code for the initialization of limma.

5.4 Experimental Setup

Division of the dataset:

The dataset was split conditionally based on model complexity to guarantee fair evaluation and prevent data leakage:

- **Linear and Shallow Models (Lasso, Ridge, and Random Forest):** Following the Scikit-survival documentation, these models used a standard 80/20 train-test split. The 80% cohort was used for model training, while the remaining 20% was reserved for final performance evaluation. In addition, five-fold cross-validation was applied during training to optimize the model hyperparameters.

- **Deep Learning Models (DeepSurv and Coxnnet):** To accommodate neural network optimization, the dataset underwent a **64/16/20 split**. Here, 64% of the data was

dedicated to core training, 16% was allocated as a validation set for hyperparameter tuning and early stopping (mitigating overfitting), and the remaining 20% served as the definitive test set to establish model credibility.

-

Grid search for Deeplearning models:

To ensure optimal performance and establish robust benchmarks for the neural network architectures (DeepSurv and Coxnnet), a systematic grid search methodology was implemented. This exhaustive search strategy was utilized to identify the most effective combination of hyperparameters, carefully balancing the networks' learning capacity with their ability to generalize. The specific hyperparameter space explored included:

- **Number of Nodes:** The total number of neurons per hidden layer. This dictates the representational capacity of the neural network, heavily influencing its ability to capture complex patterns and achieve stable convergence.
- **Dropout Rate:** A regularization technique defining the percentage of neurons that are randomly deactivated (or "dropped") during a training iteration. This forces the network to learn robust, redundant representations, effectively mitigating overfitting.
- **Batch Normalization:** A mechanism that standardizes the activations of intermediate layers by adjusting the inputs to have a mean of zero and a variance of one. This stabilizes the learning process, reduces internal covariate shift, and significantly accelerates convergence.
- **Learning Rate:** A critical optimization scalar controlling the step size at each iteration of gradient descent. It dictates how aggressively the model updates its weights, directly impacting both the speed and stability of the learning process across epochs.
- **Weight Decay:** An L2 regularization penalty applied to the network's weights. By constraining the magnitude of these weights, weight decay promotes smoother decision boundaries, making the learning process more stable and further reducing the risk of overfitting.
- **Batch Size:** The defined subset of training samples processed prior to executing a gradient update. Optimizing the batch size ensures a steady balance between the stability of the gradient estimations and overall computational efficiency

5.4 Evaluation Metrics

In these experimentations the main metrics that would be used are the:

- **The C-Index (Concordance Index)**: is a statistical metric used to evaluate the performance of predictive models, mainly those assessing time-to-event (survival analysis) or risk in the patients. It measures the model's ability to accurately rank the order of events. Its score ranges from between 0.5 to 1.0, 0.5 meaning a completely random guess and 1.0 perfect predictions.

- **The Log-Rank Test:** is a statistical hypothesis test used to compare the survival distributions of two or more independent groups. If the gene has a 0.05 or below the gene is significant, if the gene is 0.06 or more the gene is non-significant

6. Results

6.1 Comparable Models

The predictive performance of the evaluated models was assessed using two primary benchmarks: the Concordance Index (C-index) and the log-rank test, as detailed in Section 5.1. The comparative results across all tested architectures are summarized in **Table 1**.

Modelo	C-index	log_rank_p_value
Cox-nnet	**0.8750**	0.019110
DeepSurv	0.8032	0.030500
Cox Elastic Net	0.7819	0.023514
Cox Lasso	0.7713	**0.016972**
Random Forest	0.7181	0.166866

Table 2: Table of comparison between the models

As demonstrated in table 2, the deep learning architectures had superior prognostic capabilities for risk on patients. Specifically, based on the C-index benchmark, the **Cox-nnet** model achieved the highest overall accuracy (0.8750) for predicting patient risk, significantly outperforming the **DeepSurv** neural network (0.8032). Among the penalized linear baseline models, the **Cox Elastic Net** provided the most robust performance (0.7819), marginally surpassing the **Cox Lasso** (0.7713). However, the **Random Forest** model exhibited the weakest concordance (0.7181), suggesting that this tree-based ensemble approach is less optimal for accurately predicting survival risk within this specific cohort.

Furthermore, the log-rank test reinforced the idea for the clinical utility of the top performing models. With exception of the **Random Forest Model** (0.1668), all models successfully stratified and classify patients into significantly distinct risk groups (p-value < 0.05) . While the **Cox-nnet** had the highest *C-index* the most statistically significant model was **Cox lasso** (0.0169).

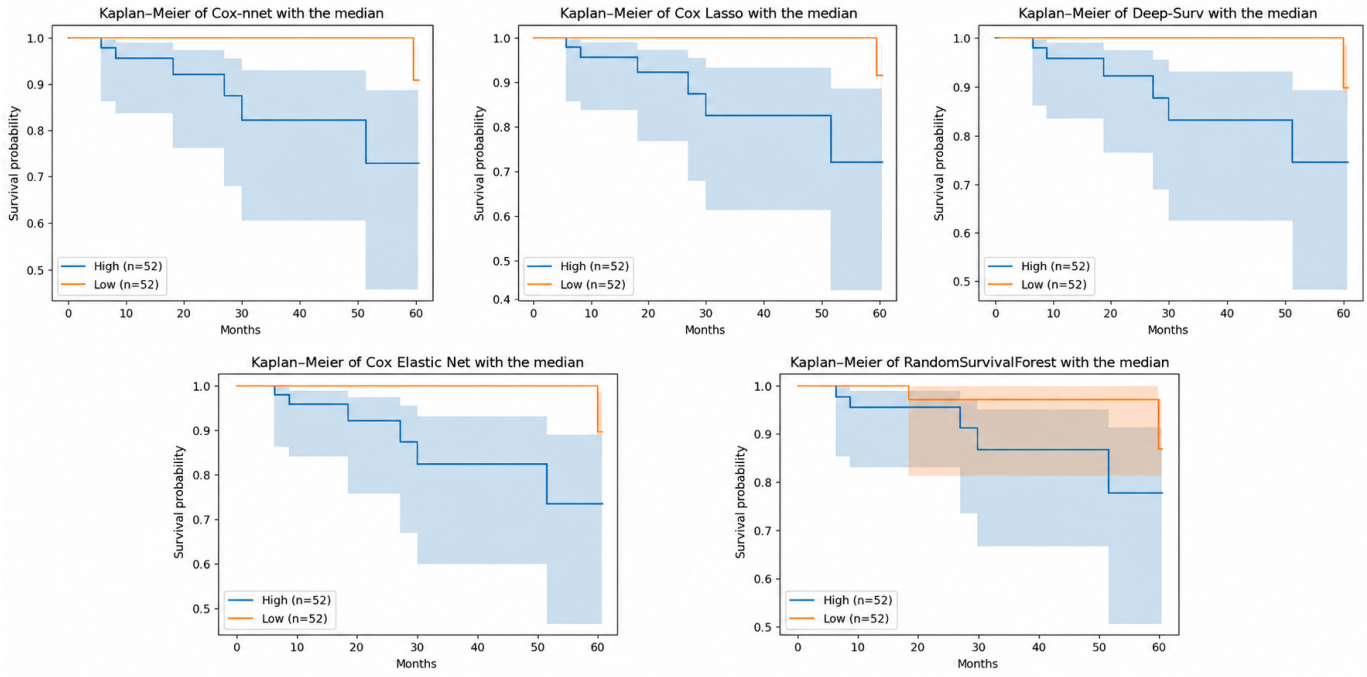


Figure 6: Kaplan-Meier of the 5 models. Risk groups are obtained using the median as a threshold.

As illustrated in **Figure 6**, Kaplan-Meier survival curves were generated for each model to visualize their ability to stratify patients into two distinct groups: high-risk patients (blue line, predictions above the median) and low-risk patients (orange line, predictions below the median). Consistent with the log-rank test results presented in **Table 1**, the top-performing models demonstrate a robust and clear separation between the survival probabilities of the high-risk and low-risk groups over the 60-month period.

In contrast, the Kaplan-Meier plot for the **Random Survival Forest** reveals a pronounced overlap in the confidence intervals (shaded regions) between the high-risk and low-risk groups. This visual overlap corroborates the model's non-significant log-rank p-value, clearly indicating that the Random Forest architecture struggles to reliably distinguish and cleanly separate the survival outcomes of the two patient groups in this dataset.

6.2 interpretability with SHAP

What is Shap?

To extract biological insights from the deep learning survival architectures that were used, DeepSurv and Cox-nnet, which are inherently non-linear "black-box" models, the SHAP (SHapley

Additive exPlanations) framework was utilized. Grounded in cooperative game theory, SHAP decomposes the model's risk prediction by assigning a specific Shapley value to each gene, treating each gene's expression level as a "player" and the model's hazard prediction as the "payout" to be explained. A gene's Shapley value is computed as the weighted average of its marginal contribution to the prediction across every possible subset of the remaining genes, comparing the model's output when the gene is included in that subset against when it is excluded. Because the number of possible feature subsets grows exponentially with the number of input genes, exact computation is intractable for high-dimensional expression data; for this reason, a permutation-based approximation was used, which estimates Shapley values by sampling a representative number of feature coalitions rather than enumerating all of them. Each gene's resulting SHAP value quantifies its signed, individual contribution to the deviation of a specific patient's predicted hazard from the baseline (average) prediction across the cohort, providing a rigorous method for gene attribution. By the additivity property of SHAP, the sum of all gene-level contributions plus this baseline value exactly reconstructs the model's output for that patient.

To ensure the statistical robustness of the genetic markers and prevent the selection of false positives (FP) arising from data sampling variability, a bootstrapping strategy was implemented (n = 50). In this pipeline, the neural networks were iteratively retrained on 50 distinct random variations of the sampled patient subsets. For each iteration, a permutation-based SHAP explainer computed the mean absolute SHAP value across the test samples for each of the 1000 limma-filtered genes; averaging the absolute values in this way yields a cohort-level importance ranking, independent of the direction of each gene's effect, which was used to rank the genes by their prognostic importance.

Results of SHAP

Selection frequency for each gene was calculated using the ratio of bootstrapping times (from a total of 50 times) that the gene appeared in the list of top 50 genes based on their average absolute SHAP value. Genes that showed a low selection frequency, meaning that they scored high in terms of SHAP importance value in few bootstrapping times, were considered statistically insignificant and thus discarded as an artifact of sampling. On the other hand, genes that showed a high selection frequency and ranked in the list of top 50 in all or nearly all bootstrapping times were considered stable and thus were retained.

Gene Symbol	DS Freq	DS Mean SHAP	CN Freq	CN Mean SHAP	Consensus
SNRPF	50	0.0284	50	0.0260	Shared
SCRN3	41	0.0208	50	0.0293	Shared
CDK7	45	0.0207	33	0.0213	DeepSurv Specific
RBMS3	50	0.0349	19	0.0208	DeepSurv Specific
OXNAD1	14	0.0181	50	0.0235	CoxNnet Specific
PHYKPL	12	0.0175	50	0.0281	CoxNnet Specific
MRPL50	10	0.0176	50	0.0254	CoxNnet Specific
GLI1	5	0.0171	50	0.0347	CoxNnet Specific
SLC35A2	50	0.0262	4	0.0189	DeepSurv Specific
TDRD7	48	0.0216	6	0.0186	DeepSurv Specific
VPS36	2	0.0148	50	0.0236	CoxNnet Specific
LCA5L	1	0.0152	50	0.0256	CoxNnet Specific
CLPB	1	0.0135	50	0.0257	CoxNnet Specific
SERPINA5	50	0.0329	0	0.0000	DeepSurv Specific
HAND2-AS1	50	0.0328	0	0.0000	DeepSurv Specific

Table 3: Table of the gene consensus of both non-linear models

To find the best prognostic genes, the frequency values of selected genes in the 50 bootstrap samples of both DeepSurv and Cox-nnet models were analyzed **Table 3**. The high level of agreement was shown for the deep learning models, and the most important genes, such as SNRPF and SCRN3, had the selection frequency equal to 100% and 91%, respectively. This fact indicates that the expression of the specified genes is crucial for the estimation of the risk probability in both models and confirms their importance for the biology of the phenomenon.

There are some model-specific prognostic genes that are selected by the different models with various frequency values. So, CDK7 gene was selected more often in DeepSurv (45/50 times) than in Cox-nnet (33/50), while OXNAD1 was selected more often in Cox-nnet (50/50) than in DeepSurv (14/50). Thus, although the models share the same purpose, the different architecture of each model allows capturing the specific transcriptomic features.

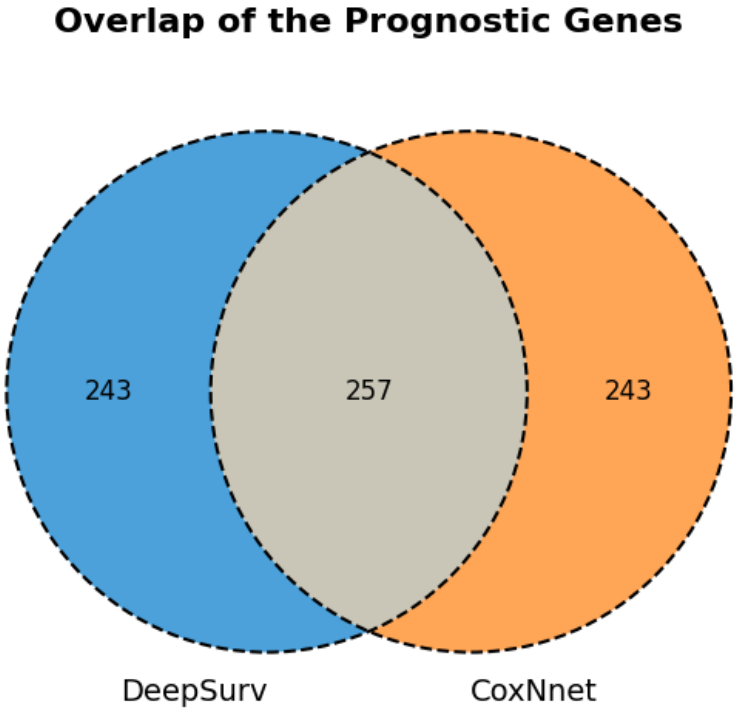


Figure 7: Venn-Diagram of the interpolation of the genes between the both non-linear models

As illustrated in Figure 7, a high degree of consensus was observed between the two deep learning models, with 257 genes shared between DeepSurv and Cox-nnet. It should be noted that the number constitutes 51.4% of the analyzed gene set (the top 500 features), demonstrating that these markers are consistently prioritized by both models for patient risk prediction. This substantial consensus suggests that these 257 genes constitute a robust, stable transcriptomic signature for survival stratification, while the remaining genes reflect complementary, architecture differences.

Comparison Category	Definition	Gene Count
Deep Learning Consensus	Genes shared by DeepSurv & Cox-nnet	**257**
— Confirmed by Linear Model	Core Consensus ∩ Elastic Net	6
— Non-Linear Drivers	Core Consensus − Elastic Net	251

Table 4: Table of the gene consensus of both non-linear models, in difference with the linear model

As shown in Table 4 below, there is a high difference in the choice of prognostic genes made by the non-linear deep learning models (DeepSurv and Cox-nnet) in comparison to the linear base model (Elastic Net). By analyzing the commonality of the 257 genes used by the neural network models in relation to the selection of genes done by the Elastic Net model, one can conclude that there are just 6 genes common to all models. On the other hand, there are 251 genes selected exclusively by the deep learning models. The above results show that deep learning models have greater ability in predicting survival stratifications due to their capability of recognizing non-linear transcriptome dependencies.

6.3 Enrichment Analysis

Finding genes is just the first step; these findings must then be linked to existing biological knowledge. Functional enrichment analysis helps by testing whether a set of selected genes is overrepresented in known biological processes, molecular functions, cellular components, or signaling pathways. This analysis often uses curated databases such as Gene Ontology and Reactome and can be performed using tools like Enrichr and gseapy. Functional enrichment analysis relies on curated annotations and prior biological knowledge assembled from multiple external sources and databases; this knowledge is organized into cellular functions, pathways with established biological effects, and, more broadly, gene sets that share a specific, well-defined function. By mapping gene lists to functions and pathways, enrichment analysis turns statistical findings into biologically meaningful hypotheses and helps interpret prognostic signatures. Notably, because a substantial proportion of the biological functions attributed to the discovered genes are related to cancer, this analysis reinforces and provides independent

support for the findings obtained in earlier stages of the pipeline. This step is therefore crucial for understanding the biological significance of the results

#	Gene_set	Term	Overlap	P-value	Adj. P-value	Odds Ratio	Combined Score	Genes
48	KEGG	Dopaminergic synapse	3/3	0.016827	0.984244	20.449902	83.532660	ADCY5; GNG12; PPP2R5B
157	KEGG	RNA transport	8/13	0.006439	0.984244	4.574422	23.079569	EEF1A1; RANGAP1; EIF4A3; SNUPN; TARDBP; ...
207	KEGG	mRNA surveillance pathway	4/6	0.040928	0.984244	5.265089	16.826912	EIF4A3; PPP2R5B; NXT2; TARDBP
325	GO BP	Cellular Response To Ketone	4/6	0.040928	0.670960	5.265089	16.826912	TNFSF4; ROCK2; ADCY5; POSTN
528	GO BP	Intracellular Protein Transport	11/20	0.004561	0.670960	3.607025	19.442558	NXT2; SURF4; ACD; TIMM44; ...
692	GO BP	Negative Regulation Of Macromolecule Biosynthesis	4/6	0.040928	0.670960	5.265089	16.826912	TIA1; CPEB3; DDX3X; EIF4A3
1073	GO BP	Protein Autoubiquitination	4/6	0.040928	0.670960	5.265089	16.826912	WWP2; RAG1; UHRF2; RNF112
1095	GO BP	Protein Localization	10/22	0.033758	0.670960	2.482667	8.412594	ACD; TIMM44; BBS4; STX16; ...
1119	GO BP	Protein Sumoylation	5/6	0.005155	0.670960	10.782178	56.798575	SAE1; RANGAP1; UBE2I; UHRF2
1145	GO BP	RNA Export From Nucleus	4/6	0.040928	0.670960	5.265089	16.826912	DDX39A; FYTTD1; NXT2; EIF4A3
1146	GO BP	RNA Metabolic Process	6/10	0.022018	0.670960	4.247184	16.206799	DDX24; DDX3X; HNRNPUL1; DICER1; ...
1242	GO BP	Regulation of CDK Activity	4/6	0.040928	0.670960	5.265089	16.826912	CDKN3; PSMD10; CDK7; CDK5RAP1
1283	GO BP	Regulation Of G1/S Transition	3/3	0.016827	0.670960	20.449902	83.532660	CDK7; DDX3X; UBE2E2
1611	GO BP	mRNA Transport	4/6	0.040928	0.670960	5.265089	16.826912	DDX39A; FYTTD1; NXT2; EIF4A3
1933	Reactome	Integration Of Energy Metabolism	4/6	0.040928	0.806388	5.265089	16.826912	ACACB; ADCY5; GNG12; STX1A
1960	Reactome	Lysine Catabolism	3/3	0.016827	0.806388	20.449902	83.532660	GCDH; ALDH7A1; PHYKPL
1984	Reactome	RNA Metabolism	17/41	0.017829	0.806388	2.136917	8.605228	SRRM2; SNUPN; DDX39A; EIF4A3; ...
2145	Reactome	RNA Pol II Termination	4/6	0.040928	0.806388	5.265089	16.826912	DDX39A; SNRPF; FYTTD1; EIF4A3
2299	Reactome	Thrombin Signaling	3/3	0.016827	0.806388	20.449902	83.532660	GNG12; TTF2; F2RL2

Table 5: Table of the enrichment analysis of the 257 genes of the two non-linear models

To investigate the biological significance of 257 consensus genes discovered via the deep learning models (**DeepSurv and Cox-nnet**), a pathway analysis was conducted **Table 5**. Out of the 257 genes, 209 genes could be annotated against the chosen reference databases (**KEGG, GO, and Reactome**), which are considered to be the gold standard databases in oncology research. The proportion of successful annotation is equal to **81.32%** of the total gene set, indicating that most of the features are known genes and therefore proving the high reliability of these genes as biological rather than statistical artifacts. After all, the results confirm that the consensus obtained using deep learning models successfully identifies the biological genes.

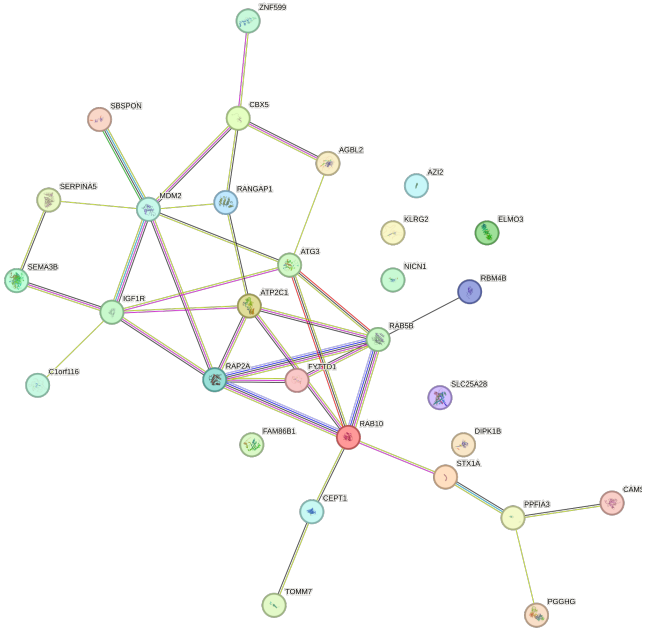
On further examination of the significantly enriched pathways ($p <= 0.05$) listed in **Table 5**, the common features of the non-linear model's consensus signature reveal their importance in terms of involvement in essential oncogenic pathways. The first type is two regulatory mechanisms, which control the process of cell cycle progression and proliferation. They are '**Regulation of G1/S Transition**' (with 3/3 genes overlapping, 100%) a necessary cell cycle control point commonly skipped in cancer due to the inhibition of tumor suppressors and DNA damage responses and '**Regulation of CDK Activity**' (with 4/6 genes overlapping, 66.67%), a pathway usually deregulated in cancers, causing uncontrolled cell proliferation, which is typical for tumor cells.

Additionally, pathways related to cell dynamics such as '**Intracellular Protein Transport**' (with 11/20 genes overlapping, 55%)—which refers to a very regulated pathway by which the cell transports the proteins to the right places—and '**Protein Localization**' (with 10/22 genes overlapping, 45.45%) show the ability of the models to detect the complexity of cellular machinery needed for tumor microenvironment remodeling. Finally, the abundance of 'RNA Transport' (with 8/13 genes overlapping, 61.53%) and '**RNA Metabolic Process**' (with 6/10 genes overlapping, 60%) shows the priority of the genes associated with transcriptional dysregulation.

6.4 Interaction between genes

Interaction between genes with Random Survival Forest and minimal depth criterion

These results present the predictive findings from the Random Survival Forest (RSF) analysis, a non-parametric machine learning approach utilized to identify key breast cancer survival genes. To select the most critical predictors from the genomic dataset, feature selection was conducted using the minimal depth criterion. In the context of decision trees, minimal depth provides a highly interpretable measure of variable importance. Specifically, it calculates the shortest distance from the root node to the first instance where a particular gene is used to split the data. Because the algorithm prioritizes the most informative variables early in the tree-building process to maximize survival differentiation, genes with a lower minimal depth are recognized as the strongest predictors. Finally, Figure 8 details the core predictive genes identified through this methodology, mapping their known, experimentally validated functional interactions to highlight their biological relevance to cancer progression.



Index	Gen i	Gen j	Interaction (Minimal Depth)
0	RAB10	STX1A	2.800
1	SBSPON	STX1A	3.067
2	ATP2C1	STX1A	3.067
3	ATP2C1	RAB10	3.067
4	RBM4B	RAB10	3.100
5	STX1A	CBX5	3.150
6	SBSPON	RAB10	3.150
7	RAB10	CBX5	3.150
8	ATP2C1	CBX5	3.167
9	FYTTD1	STX1A	3.200
10	FYTTD1	RAB10	3.200
11	ELMO3	STX1A	3.233
12	ELMO3	RAB10	3.233
13	RAB10	CAMSAP1	3.233
14	SBSPON	FAM69B	3.250
15	STX1A	FAM69B	3.275
16	RAB10	FAM69B	3.275
17	ATP2C1	FYTTD1	3.300
18	ELMO3	CBX5	3.333
19	SEMA3B	STX1A	3.338
20	ATHL1	ATP2C1	3.350
21	RAP2A	STX1A	3.400
22	RBM4B	STX1A	3.400
23	RAP2A	RAB10	3.400
24	FYTTD1	SLC25A28	3.425
25	ELMO3	FAM69B	3.458
26	RANGAP1	STX1A	3.483
27	RANGAP1	RAB10	3.483
28	ATHL1	STX1A	3.500
30	KLRG2	STX1A	3.500
31	ELMO3	ATP2C1	3.500
32	SLC25A28	RAB10	3.525
33	SLC25A28	STX1A	3.525
34	AGBL2	RAB10	3.525
35	SEMA3B	RAB10	3.525

Figure 8 and Table 5: Graph of the STRING and table from the interaction between genes using Random Survival Forest and Minimal Depth

To validate the biological relevance of the interactions identified by our machine learning model, the results were cross-referenced with the STRING database a widely used bioinformatics resource that aggregates known and predicted protein-protein interactions (both physical and functional) derived from experimental data, computational predictions, and literature text-mining. By mapping the top predictive gene pairs from the RSF model onto the STRING network, we can determine whether the statistically driven survival predictors correspond to real, established biological pathways.

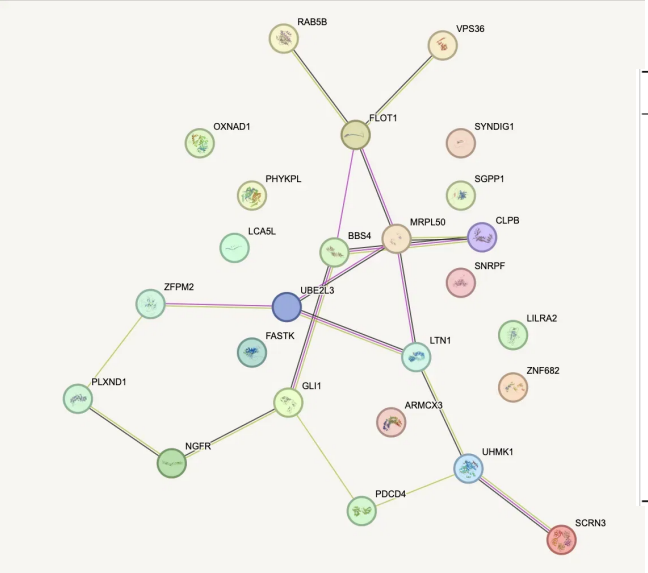
The core interactions identified through the RSF minimal depth analysis and supported by the STRING network include:

The most powerful gene pair found by our model is the pair between RAB10 and STX1A (Minimal Depth = 2.800), which is also supported by STRING based on the expression and databases of biological information. In the same way, ATP2C1 proves itself to be an important gene since it demonstrates interactions with both STX1A and RAB10 (Minimal Depth = 3.067), corresponding to the key nodes of the STRING network.

Furthermore, SBSPON exhibits predicted interactions with STX1A (Minimal Depth = 3.067) and RAB10 (Minimal Depth = 3.150) and is also connected to the central hub in the STRING graph. The gene ELMO3 reveals predicted interactions with both STX1A and RAB10 (Minimal Depth = 3.233). Moreover, FYTTD1 demonstrates predicted interactions with STX1A and RAB10 (Minimal Depth = 3.200), with STRING also illustrating the connectivity to the central cluster. Finally, RANGAP1 appears in both algorithms, predicting interactions with STX1A and RAB10 (Minimal Depth = 3.483) and appearing as a connected node in STRING.

As can be seen from the figure and table, the correlation between predictions of the Random Survival Forest and the interaction network of the STRING database indicates that the model finds biologically significant associations between cancer-related genes rather than random noise. Therefore, such results prove the biological validation of the Minimal Depth measure of identifying gene-gene interactions related to breast cancer survival.

Interaction between genes with Cox-Nnet



Index	Gene i	Gene j	Interaction	z-score	p-value
0	SCRN3	ZNF682	Synergy (+)	+2.971	**0.0015**
1	FLOT1	MRPL50	Synergy (+)	+2.223	**0.013**
2	GLI1	MRPL50	Synergy (+)	+1.756	**0.040**
3	NGFR	VPS36	Synergy (+)	+1.682	**0.046**
4	BBS4	MRPL50	Synergy (+)	+1.557	0.060
5	FLOT1	LTN1	Synergy (+)	+1.546	0.061
6	FLOT1	NGFR	Synergy (+)	+1.535	0.062
7	MRPL50	LTN1	Synergy (+)	+1.270	0.102
8	UBE2L3	MRPL50	Antagonism (−)	−5.215	1.000
9	UBE2L3	FASTK	Antagonism (−)	−5.215	1.000
10	ARMCX3	UHMK1	Antagonism (−)	−6.513	1.000
11	UHMK1	SCRN3	Antagonism (−)	−4.289	1.000

Figure 9 and Table 6: Graph of the STRING and table from the interaction between genes using Cox-Nnet

To compare between the two algorithms Random Survival Forest and Cox-Nnet it has been done with the highest-ranked gene-gene interactions used by the Cox-Nnet algorithm, the results obtained have been validated using the STRING protein-protein interaction database. As illustrated in Figure 9 and Table 6, there are several interactions predicted by the model that are confirmed by the STRING database, demonstrating that the model discovers biologically reasonable connections.

The most significant pair of genes predicted by the Cox-Nnet is the interaction between SCRN3 and ZNF682(With a z-score=2.971, p-value=0.0015), which represent the strongest connection in the permutation analysis. This pair of genes are supported by STRING through co-expression and evidence, suggesting that his pair is relevant to cancer survival. In the same way FLOT1 proves to be a central gene, having a significant interactions with both MRPL50 (With a z-score=2.223, p-value=0.013) and functioning as a bridge in the STRING network with connections to RAB5B, VPS36, BBS4 and LTN1.

Furthermore GLI1 reveals a significant predicted interaction with MRPL50 (With a z-score=1.756, p-value=0.040), a connection also reflected in the STRING network where GLI1 appears as a part of the central cluster around UBE2L3. NGFR demonstrated a significant interaction with VPS36 (With a z-score=2.971, p-value=0.0015), with the STRING illustrating the connection of NGFR to PLXND1 and ZFPM2 network.

Finally the dominant network in the STRING graph are these genes, MRPL50, BBS4, FASTK, GLI1 and LTN1, all of which also participate in the Cox-Nnet further reinforcing the biological coherence of the predictions .

In conclusion, as can be seen from the figure and the table, the concordance between the predictions of Cox-Nnnet and the interaction network of the STRING database indicates that the model identifies biologically significant associations between cancer-related genes rather than random noise. Therefore, these results support the biological validity of the interaction scores derived from the Cox-Nnet neuronal network as a measure for identifying interactions related to cancer survival.

7. Conclusions and Future work

This investigation successfully demonstrates the synergy between advanced machine learning and translational bioinformatics to enhance prognostic stratification in breast cancer. A primary conclusion of this investigation is that non-linear deep learning models like **Cox-Nnet** and **DeepSurv** tend to be a superior choice for the risk stratification of patient groups and the prediction of complex prognostic interactions among genes, consistently outperforming traditional linear models. By capturing intricate, high-dimensional transcriptomic dynamics that conventional methods obscure, these architectures provide a significantly more accurate representation of patient risk.

Furthermore, the computational robustness of these non-linear models was biologically validated through protein-protein interaction network mapping in STRING. Rather than isolating statistical artifacts, the algorithms prioritized highly cohesive molecular clusters, highlighting key regulatory drivers in RNA transport and vesicle-mediated trafficking, such as RANGAP1 and FYTTD1. While data constraints from cBioPortal notably the exclusion of unclassified molecular subtypes highlight the need for broader multi-omic validation, this project establishes a robust computational framework. Ultimately, these findings bridge the gap between data science and oncology, offering a mathematically sound, biologically validated pipeline capable of discovering novel prognostic signatures to advance personalized therapeutic strategies.

One key direction for future research is validation of these models in a much larger and more balanced cohort of patients. The dataset used in this research, collected via cBioPortal, was restricted due to both small initial size (819 patients) and a certain clinical limitation: 37% (n = 299) of the patients were labeled as belonging to "UNKNOWN" molecular subtype. As these samples cannot be easily stratified into specific subtypes of breast cancer, they were discarded, leaving only 519 patients available for analysis. Larger clinical registry would help mitigate data loss and provide more representative sample.

However, expansion of the dataset will demand higher computation resources that in turn would allow more advanced phases of optimization of machine learning algorithms. Due to higher computing capacity, further studies will be able to perform exhaustive optimization of deep learning algorithms, tuning various hyperparameters of hidden layers and neuron density as well as allowing Random Survival Forests to create deeper trees, leading to better understanding of non-linear gene-gene interaction and increased robustness of model. Finally, translation of computational signature in translational oncology, via collaboration with clinicians to test these gene targets in vitro or using patient biopsies is a crucial task ahead of us.

Conclusions and Future Work

Along with expansion of grid-search of neural network architectures, future research will have to find the right balance between computational complexity and additional value provided by deep learning algorithms. Comparative profiling of training performance and memory usage of various deep learning layers versus Random Survival Forest will guarantee that future computational tool will be easy to use on regular computer hardware without dedicated GPU cluster.

Explainable AI (XAI) and clinical translation: while minimal depth criterion helped to identify key predictors, integration of XAI such as SHAP (SHapley Additive exPlanations) framework is vital to address deep learning model opacity. Ultimately, collaboration with clinical researchers to validate these computational signatures in vitro remains the key next step to connect machine learning and real oncology.

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